

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: *L. reuteri* erythromycin resistance plasmid, pLUL631 (10.2kB, covalently closed circular form) and derivatives.

Species Used *Lactobacillus reuteri*, DSM 20016

Before the Pulse

Cell Growth Medium *Lactobacillus* Carrying Medium (LCM) + 40 mM glucose

Growth Phase at Harvest O.D. (600) = 0.5 to 1.0 (mid-log)

Pre-pulse Incubation Ice, 1 to 3 min

Wash Solution Distilled water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 30% PEG 1500 /distilled water

Cuvette Gap 0.2 cm

Cell Density Concentrated culture ~100X:

Voltage 2.0 to 2.5 kV

Volume of Cells 100 µl

Field Strength 10.0 to 12.5 kV/cm

DNA Concentration 0.001 to 5 µg

DNA Resuspension Buffer TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0) or water

Capacitor 25 µF

Volume of DNA < 10µl

Resistor (Pulse Controller) 200 Ω

After the Pulse

Time Constant 0.5 msec

Outgrowth Medium LCM + 40 mM glucose

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Plasmid, **20**:171-174.
 Electroporation: Axelsson, L.T. & Ahrne, S.E.I. 1990. Transformation of *Lactobacillus reuteri* with electroporation: studies on the erythromycin resistance plasmid pLUL631. In: Axelsson, L. *Lactobacillus reuteri*, a member of the gut bacterial flora. Dissertation. Report 44. Dept. Microbiology, Swedish Univ., Uppsala, Sweden
 Ahrne, S., & Axelsson, L. 1990. FEMS Microbiology. Rev. Abstract-A13, **87**: 12.

Outgrowth Temperature 37 °C

Length of Incubation 1.5 hrs

Selection Method or Assay Used erythromycin or chloramphenicol resistance, 10 µg / ml

Electroporation Efficiency 10 (7) to 10 (8) transformants / µg DNA

Per Cent Survival Not known

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Survey Number

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